# TRANSISTOR

## THEORY OF OPERATION

*prepared by:*

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Electronics Engineer

# GRADING SYSTEM

# FINAL GRADE

30%

PRELIM

30%

MIDTERM

**40%**

ENDTERM

# TERM GRADE

50%

WEEKLY

50%

EXAM

# 100%

LABORATORY

# PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

# TOPIC OUTLINE

## Transistor Construction and Operation

## Transistor Configurations

- Common Base
- Common Collector
- Common Emitter

## Characteristic Curve

## Regions of Operation

## Reading Datasheet

# TRANSISTOR CONSTRUCTION AND OPERATION

# THE FIRST TRANSISTOR

December 23, 1947

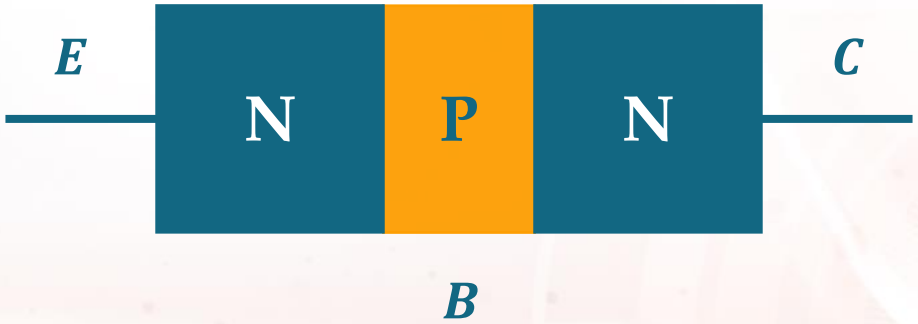
Dr. William Shockley

Dr. John Bardeen

Dr. Walter Brattain

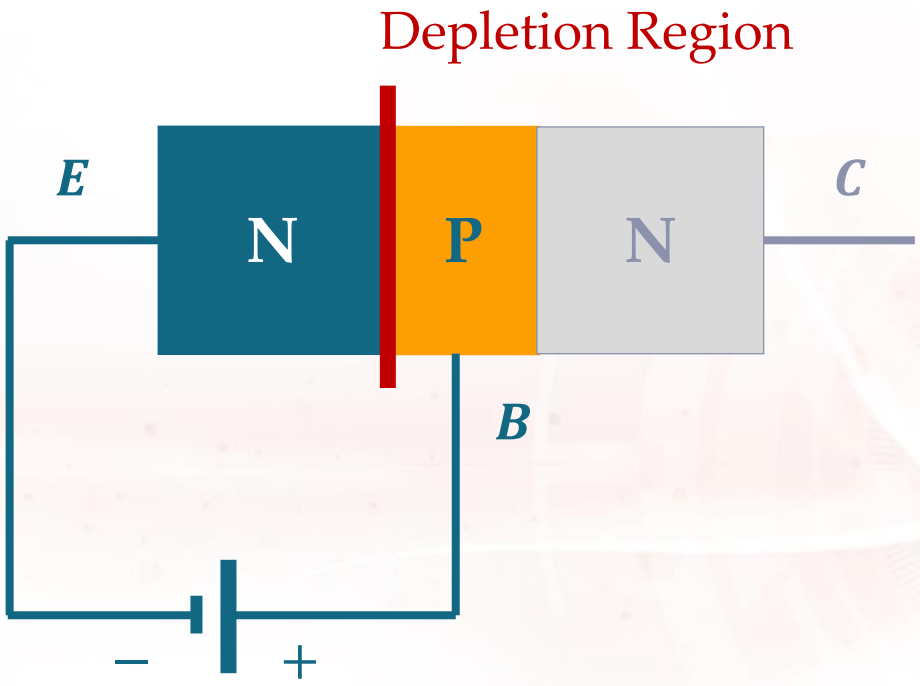
# CONSTRUCTION

## NPN



## PNP

# EMITTER-BASE DIODE

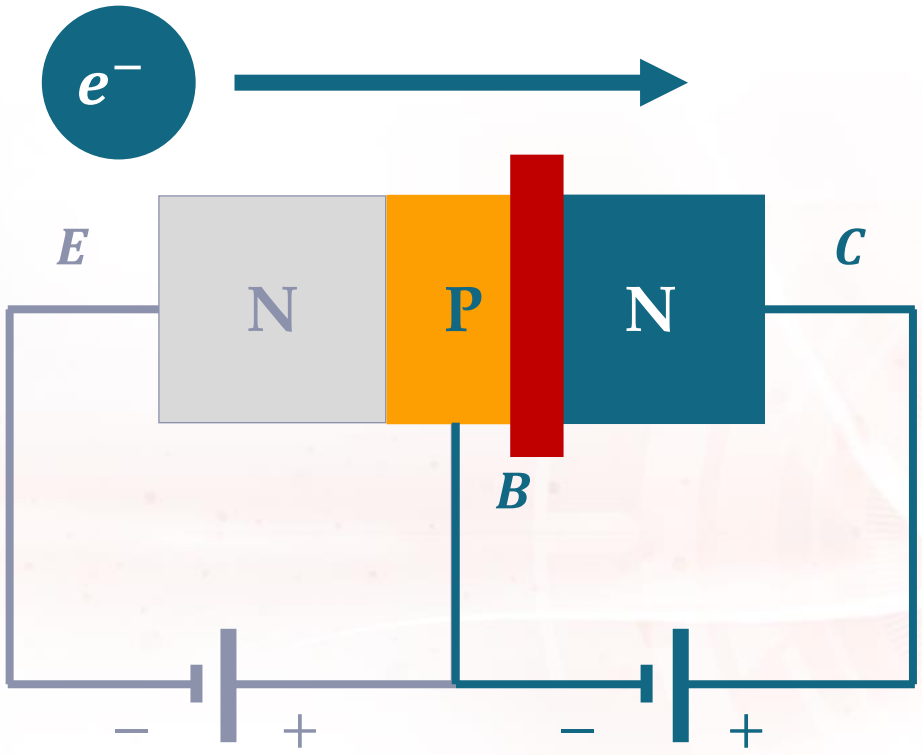



**forward**-biased

Emitter is **heavily** doped

Base is **lightly** doped

# COLLECTOR-BASE DIODE

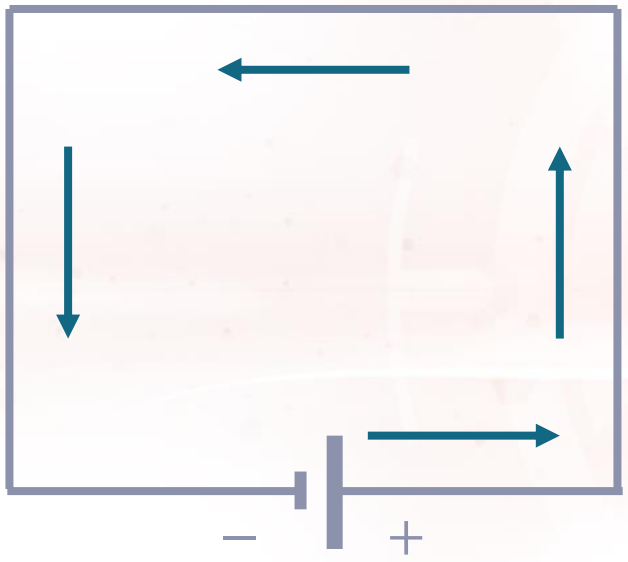



**reverse**-biased

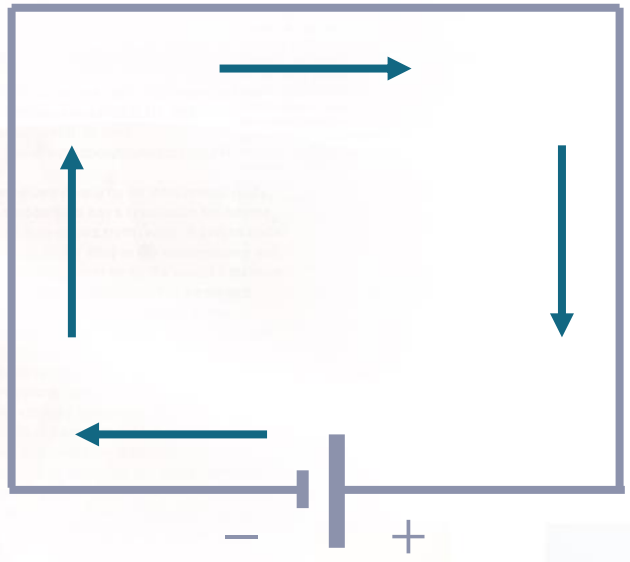
Collector is doped at an **intermediate** level

# CURRENT FLOW NOTATION

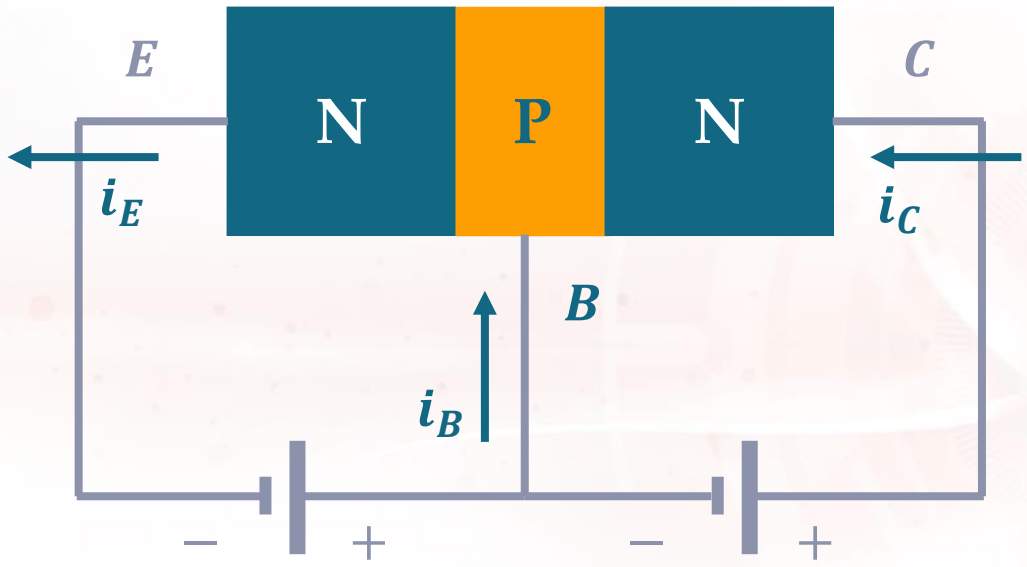
## Conventional Flow Notation




## Electron Flow Notation



# KIRCHHOFF'S CURRENT LAW



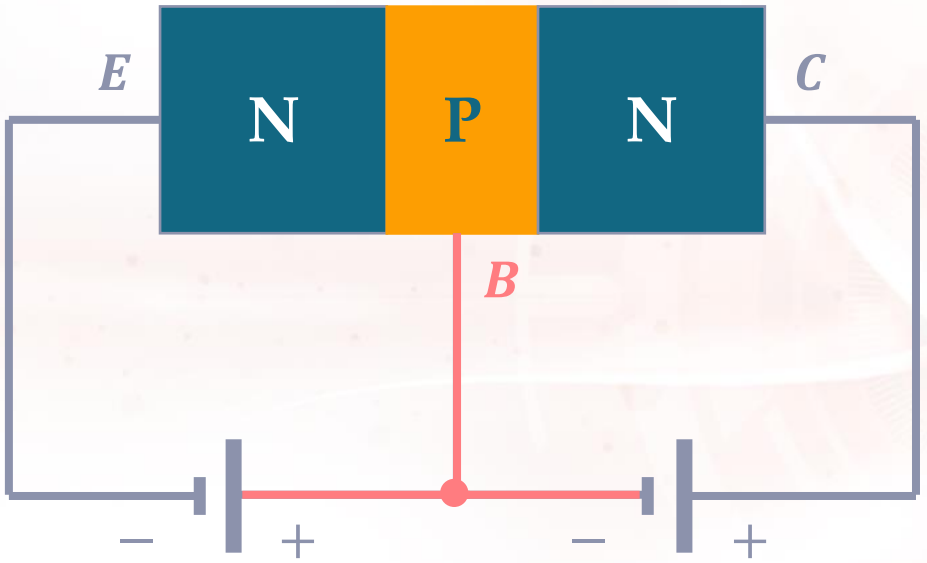

KCL @ B,

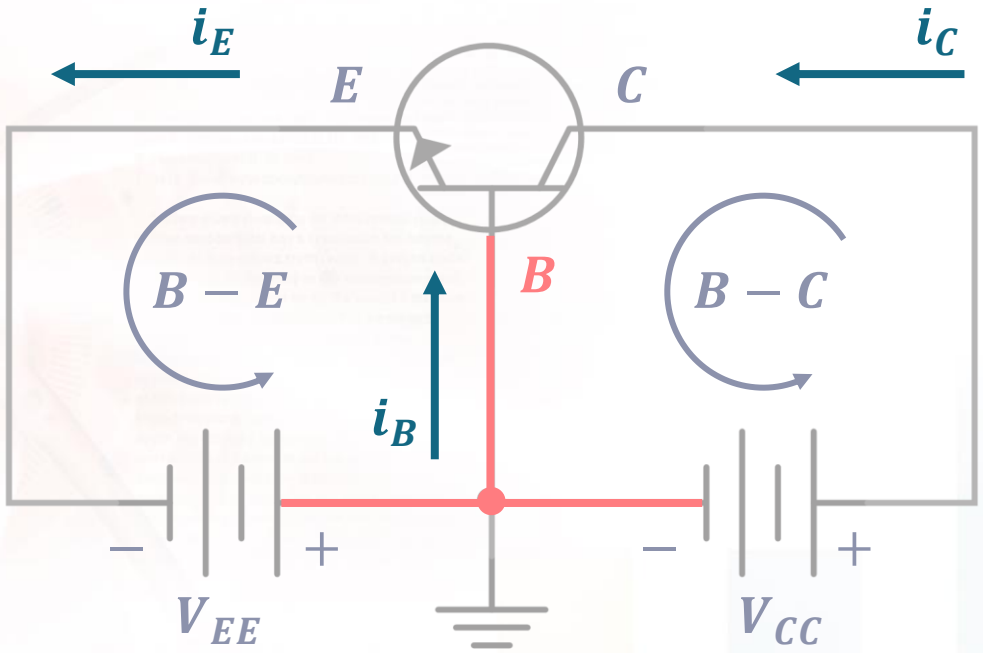
$$-i_C - i_B + i_E = 0$$

$$i_E = i_C + i_B$$

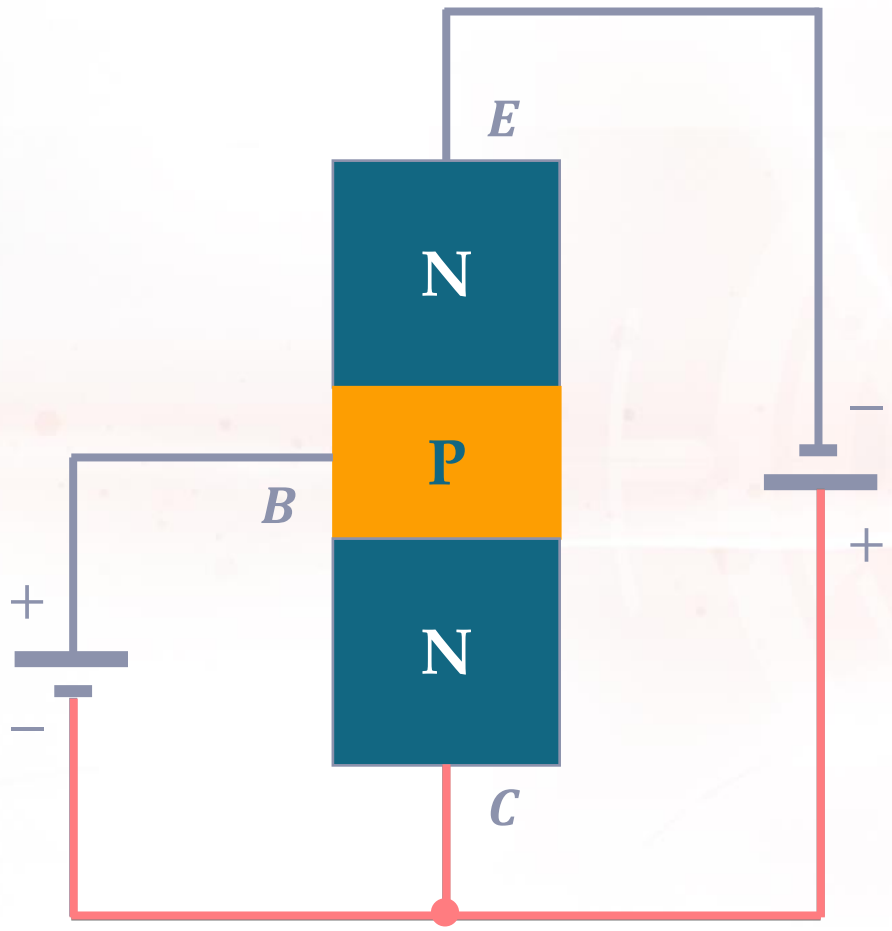
# TRANSISTOR CONFIGURATIONS

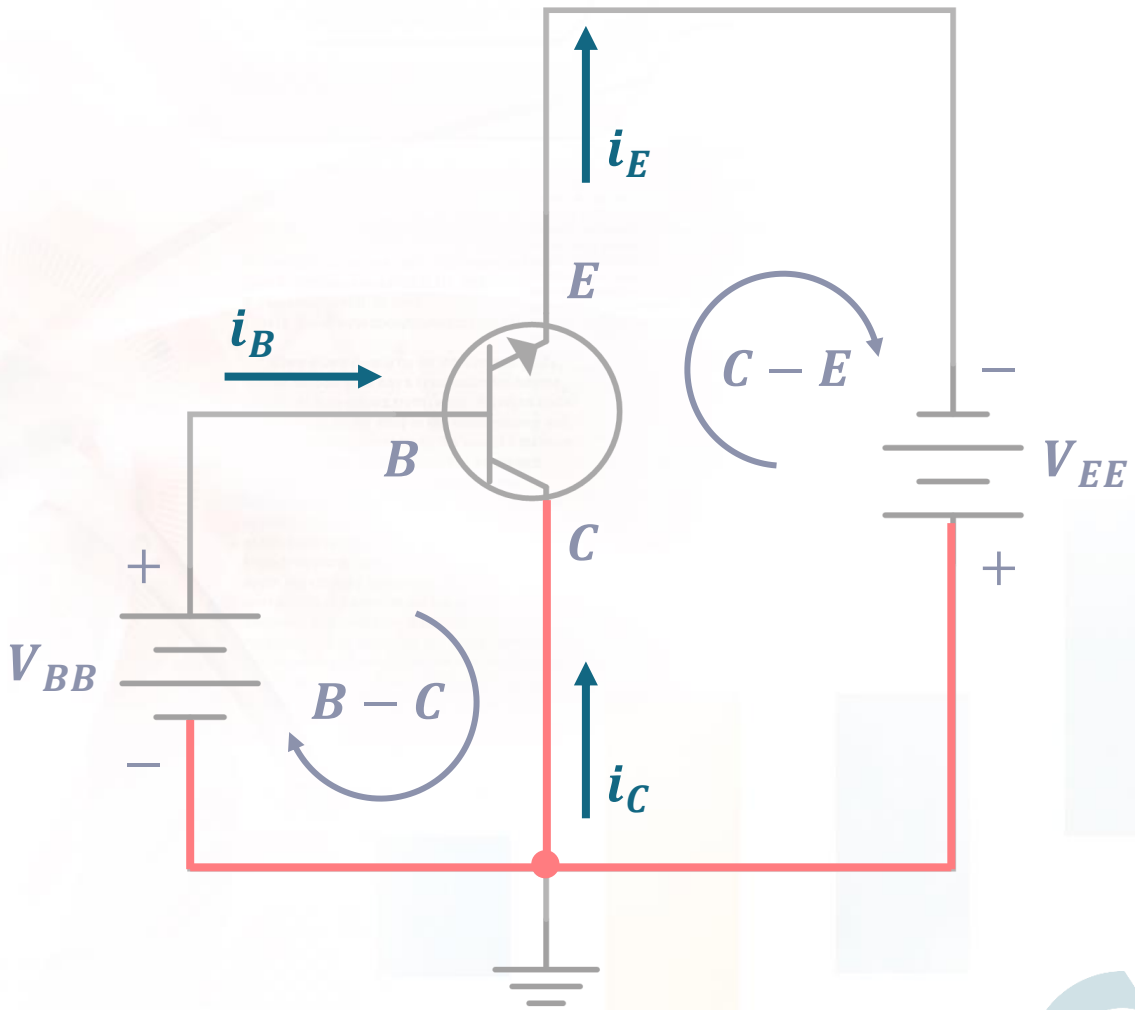
# COMMON-BASE



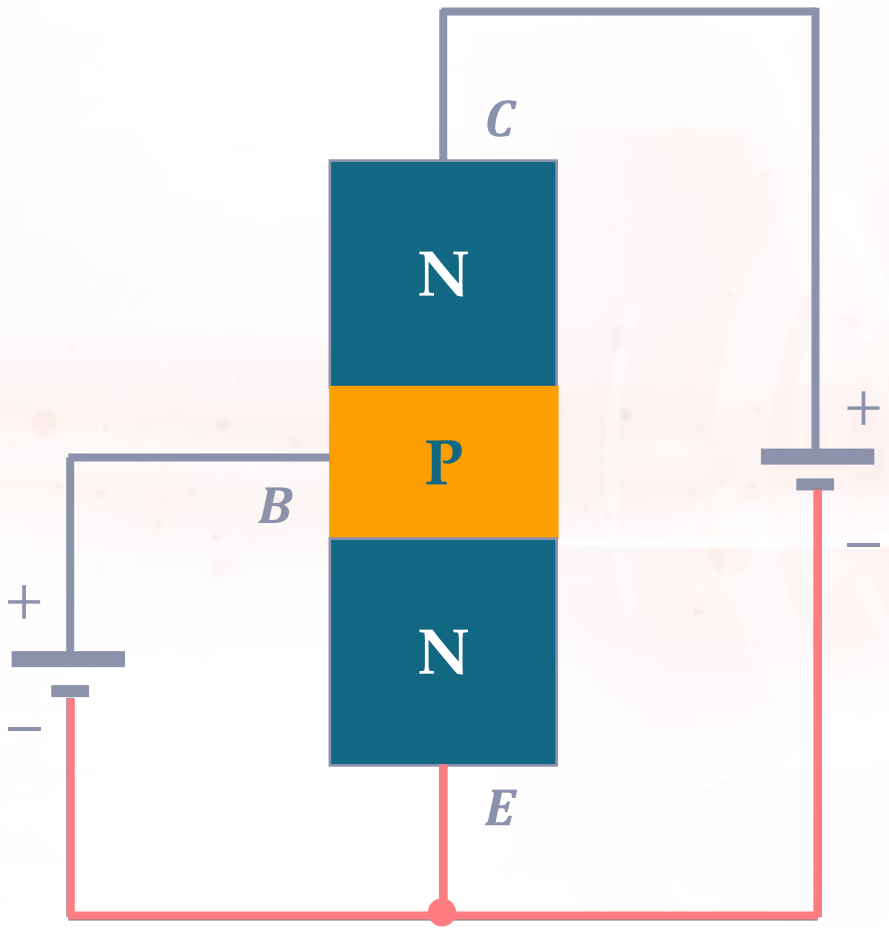



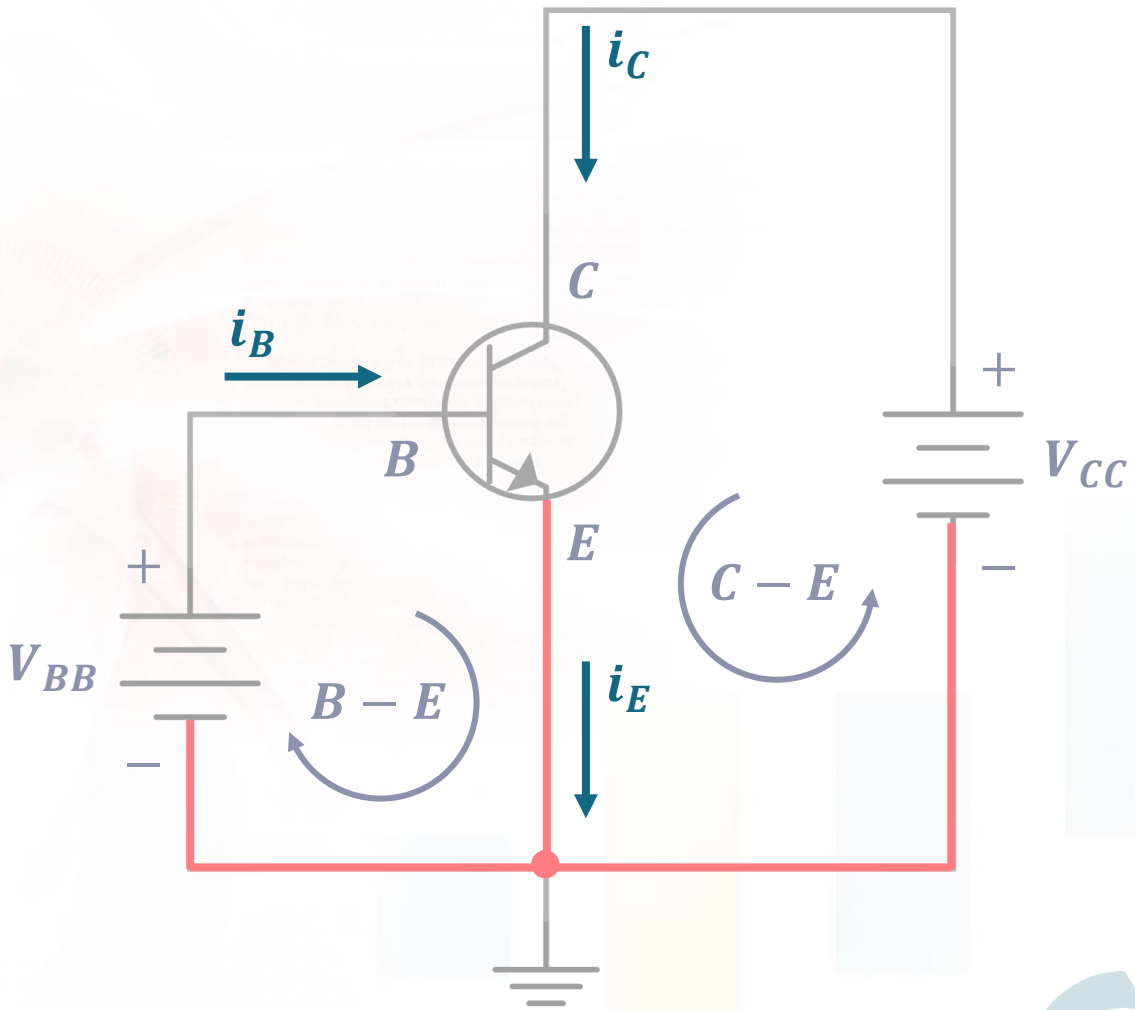
# COMMON-COLLECTOR



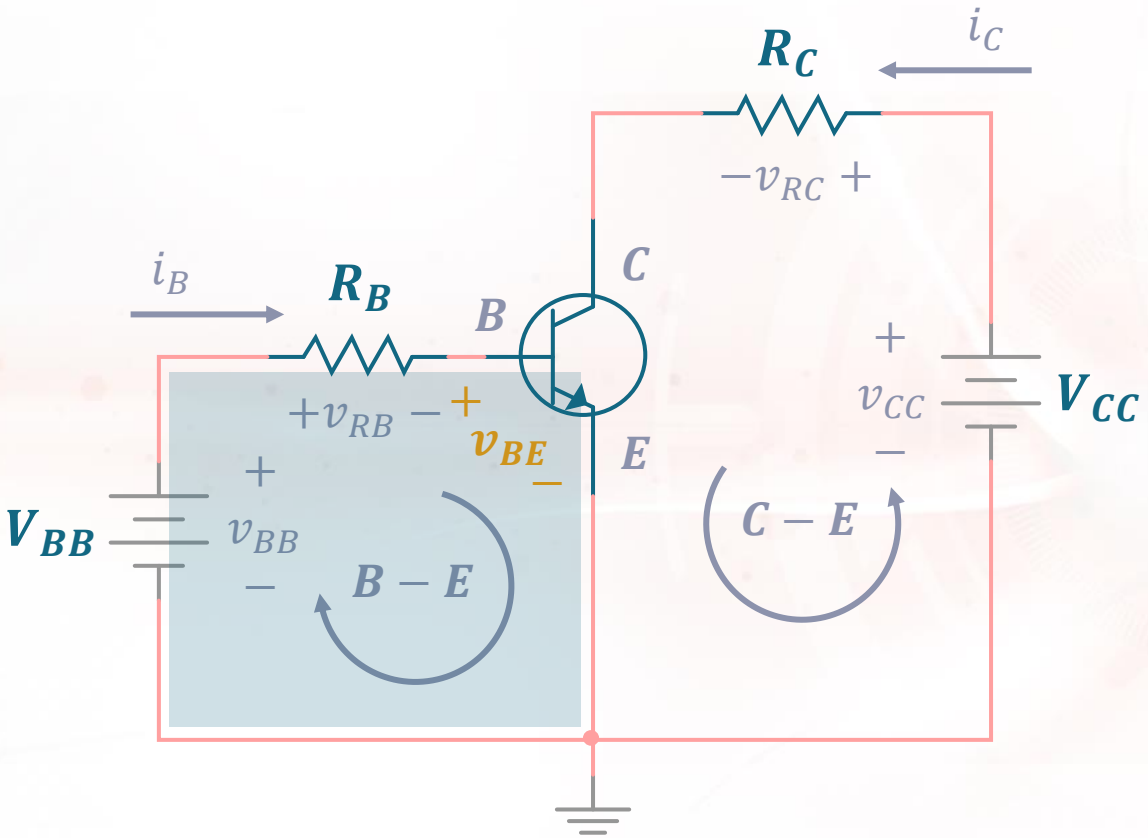
# COMMON-EMITTER

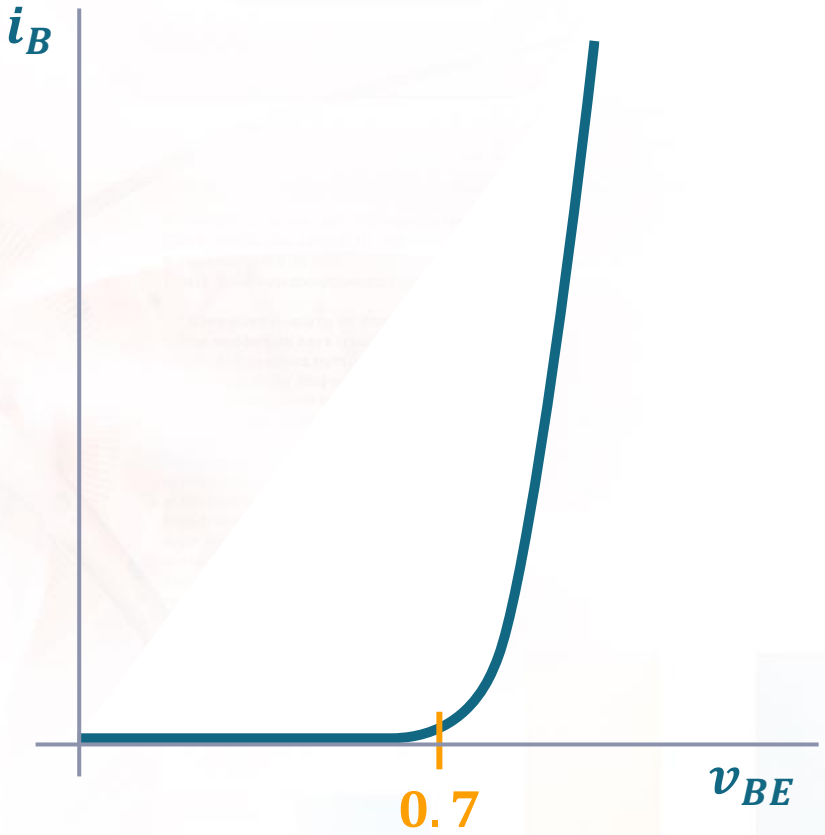


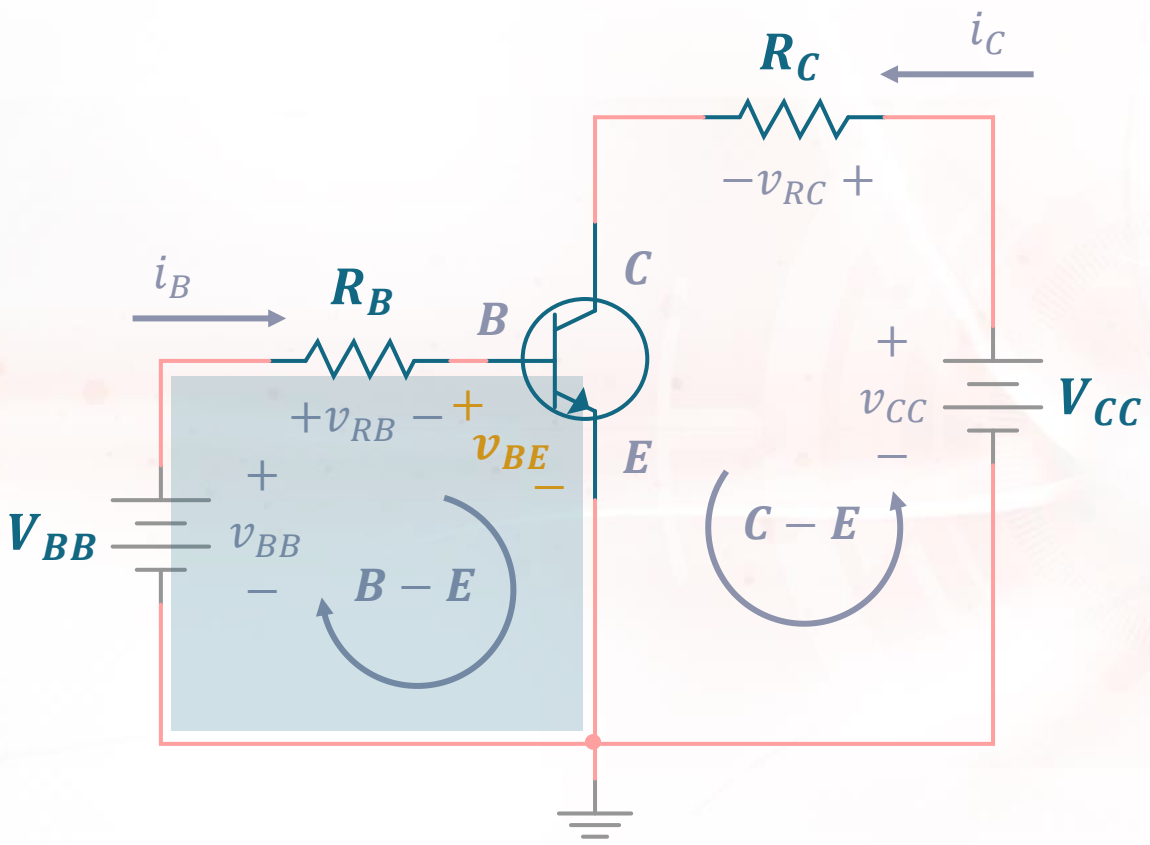
# CHARACTERISTIC CURVE

# THE BASE CURVE



# BASE CURRENT




Base-Emitter Loop,

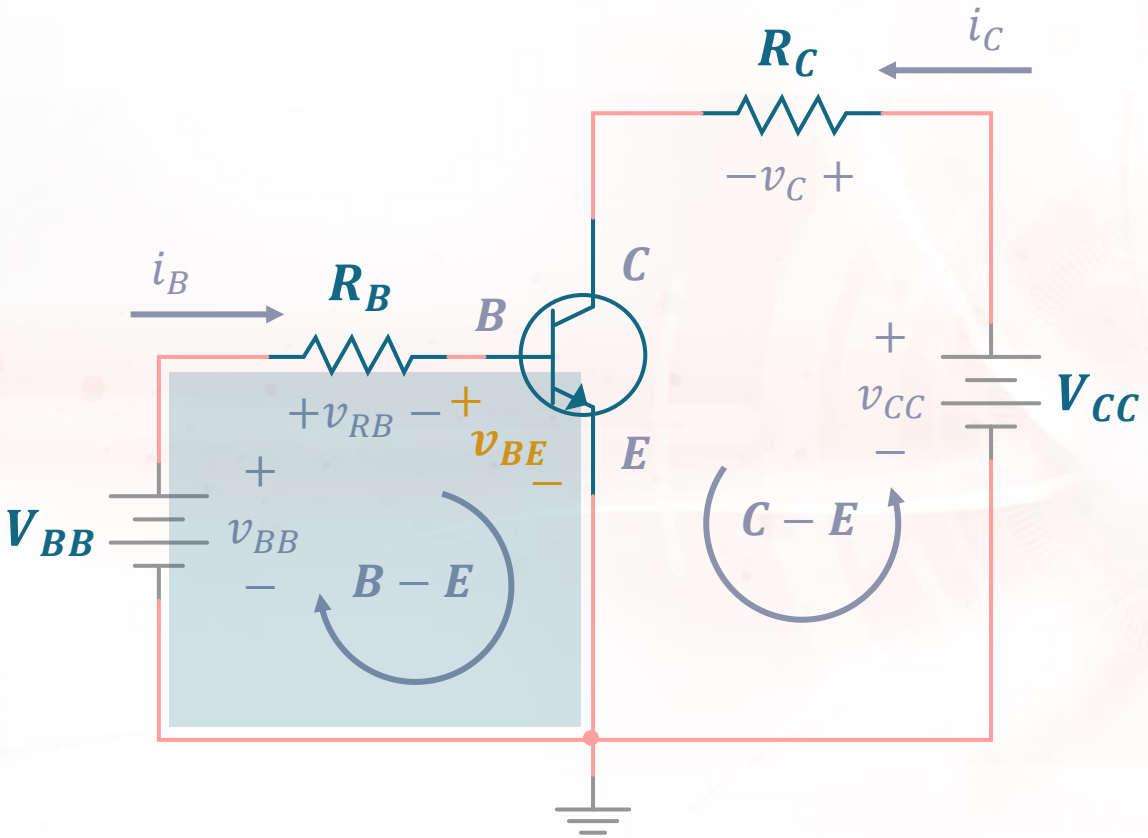
$$-v_{BB} + v_{RB} + v_{BE} = 0$$

$$v_{RB} = v_{BB} - v_{BE}$$

$$i_B R_B = v_{BB} - v_{BE}$$

$$i_B = \frac{V_{BB} - v_{BE}}{R_B}$$

# BASE CURRENT

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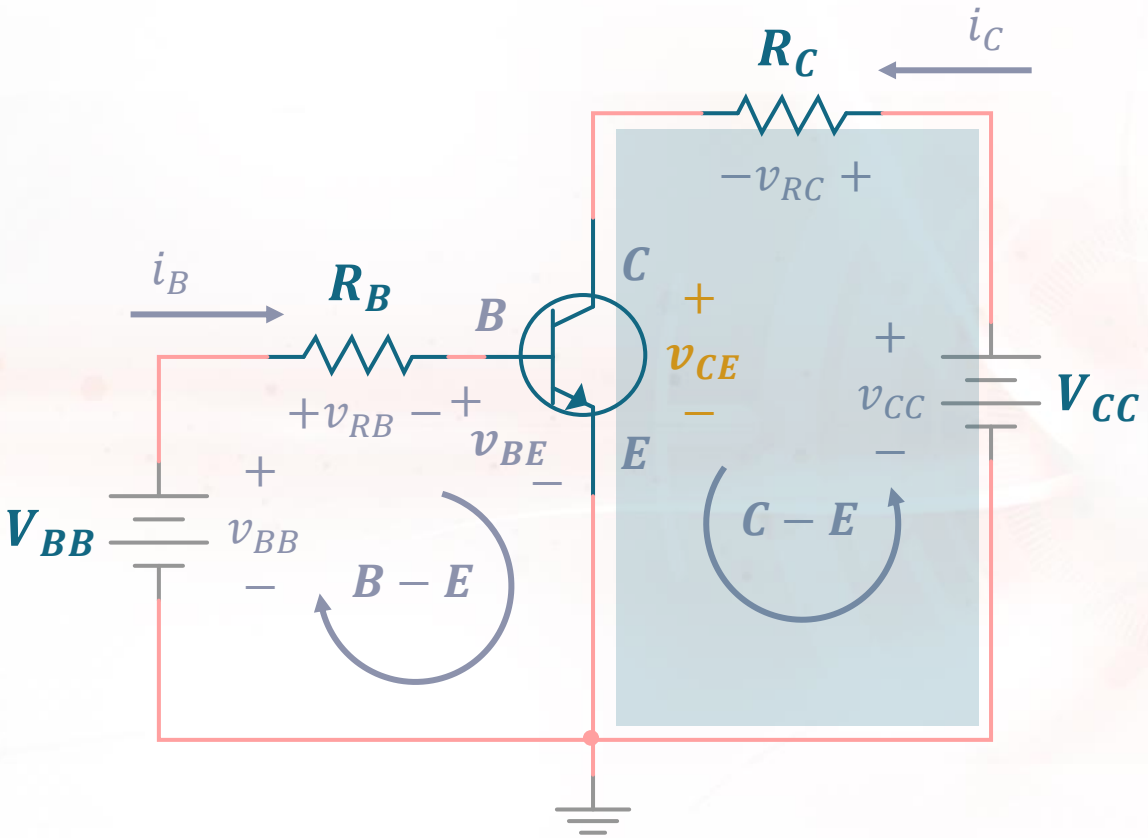
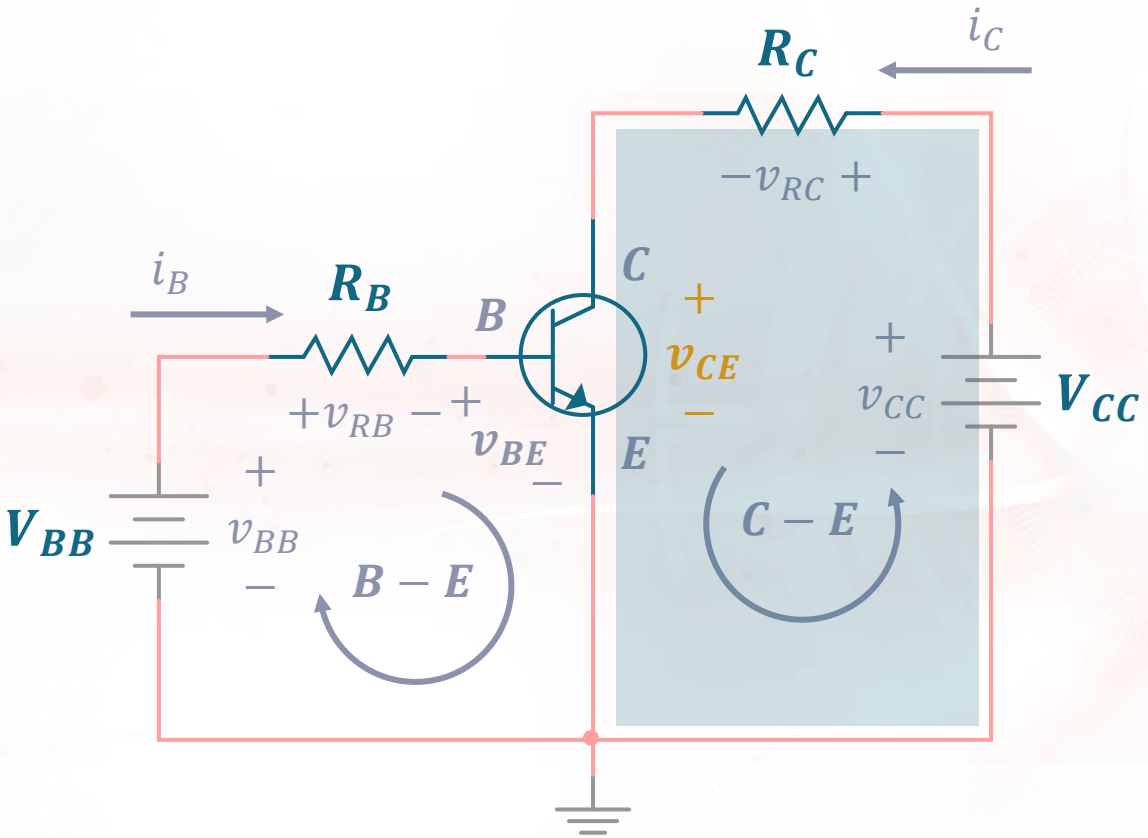
CE

# THE COLLECTOR CURVE



# COLLECTOR-EMITTER VOLTAGE



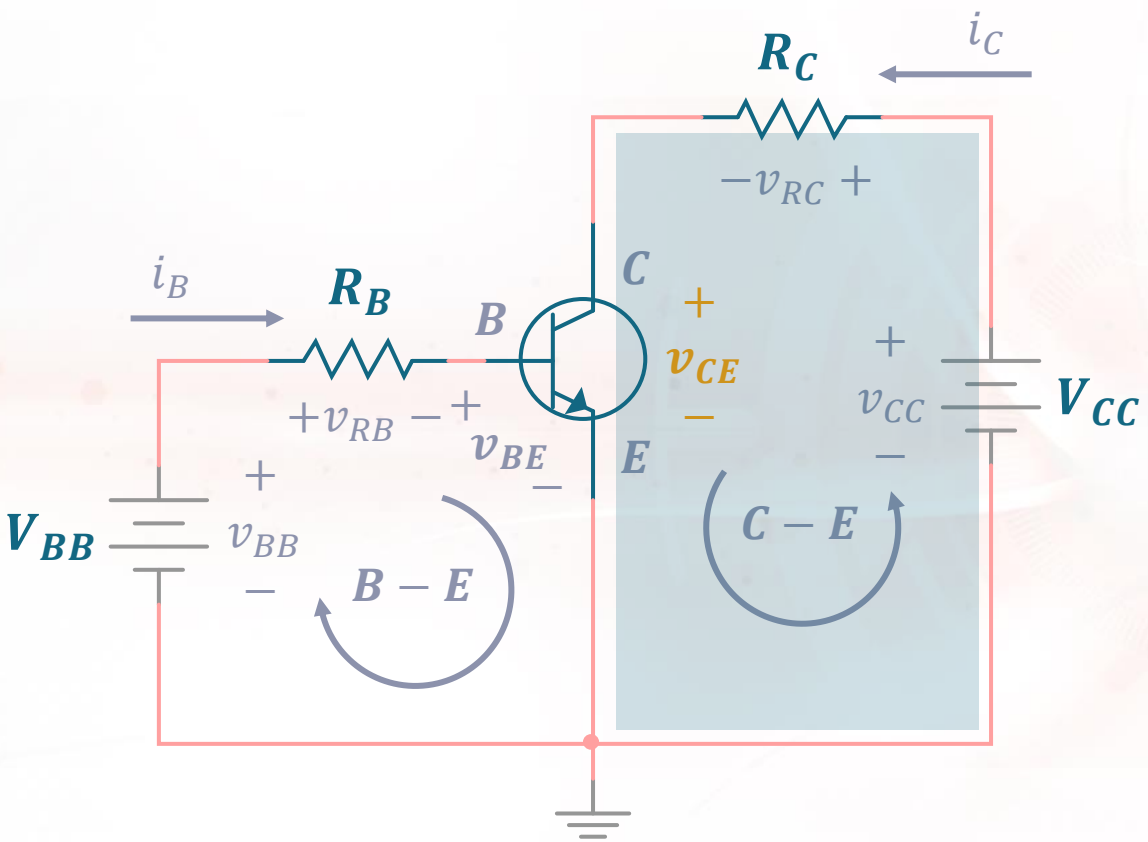

Collector-Emitter Loop,

$$-v_{CE} - v_{RC} + v_{CC} = 0$$

$$v_{CE} = v_{CC} - v_{RC}$$

$$v_{CE} = V_{CC} - i_C R_C$$

# COLLECTOR-EMITTER VOLTAGE



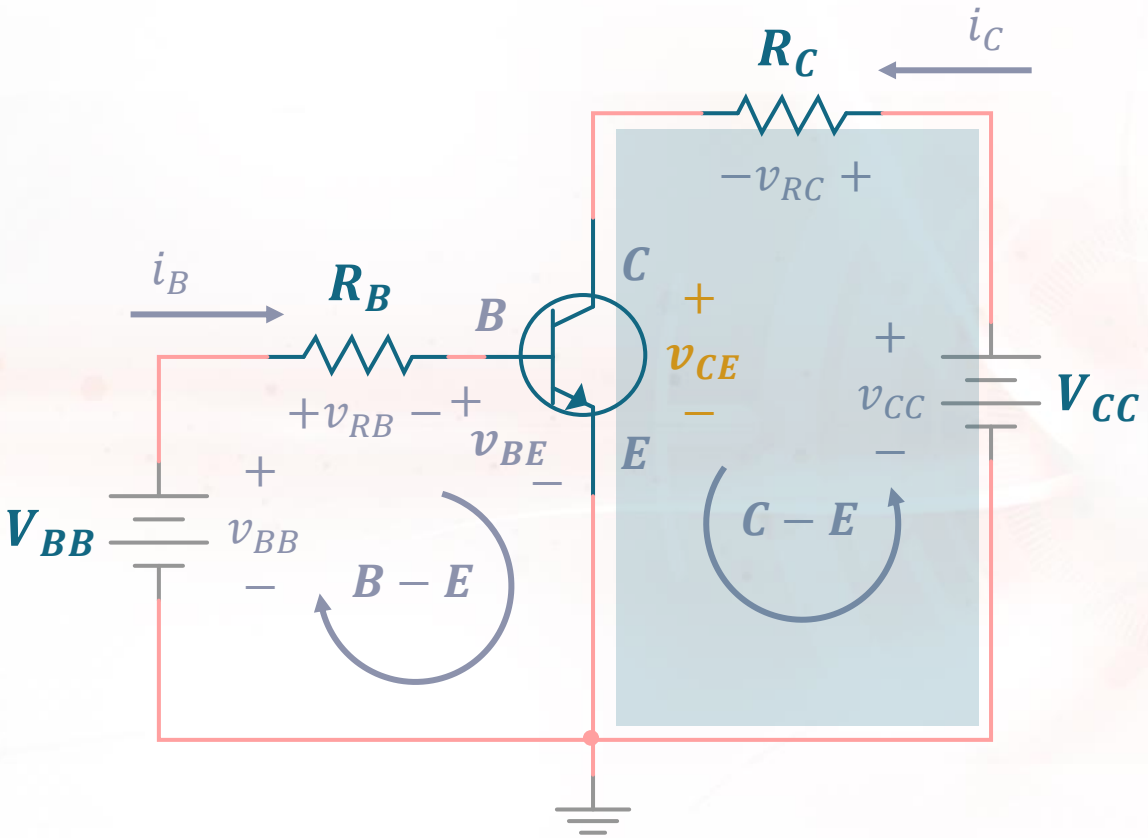

Collector-Emitter Loop,

$$-v_{CE} - v_{RC} + v_{CC} = 0$$

$$v_{CE} = v_{CC} - v_{RC}$$

$$v_{CE} = V_{CC} - i_C R_C \Big|_{CE}$$

# COLLECTOR-EMITTER VOLTAGE




Collector-Emitter Loop,

$$-v_{CE} - v_{RC} + v_{CC} = 0$$

$$v_{CE} = v_{CC} - v_{RC}$$

$$v_{CE} = V_{CC} - i_C R_C \Big|_{CE}$$

# CURRENT GAIN

## Alpha

$$\alpha_{dc} = \frac{i_C}{i_E}$$

$$\alpha_{dc} \approx 1$$

## Beta

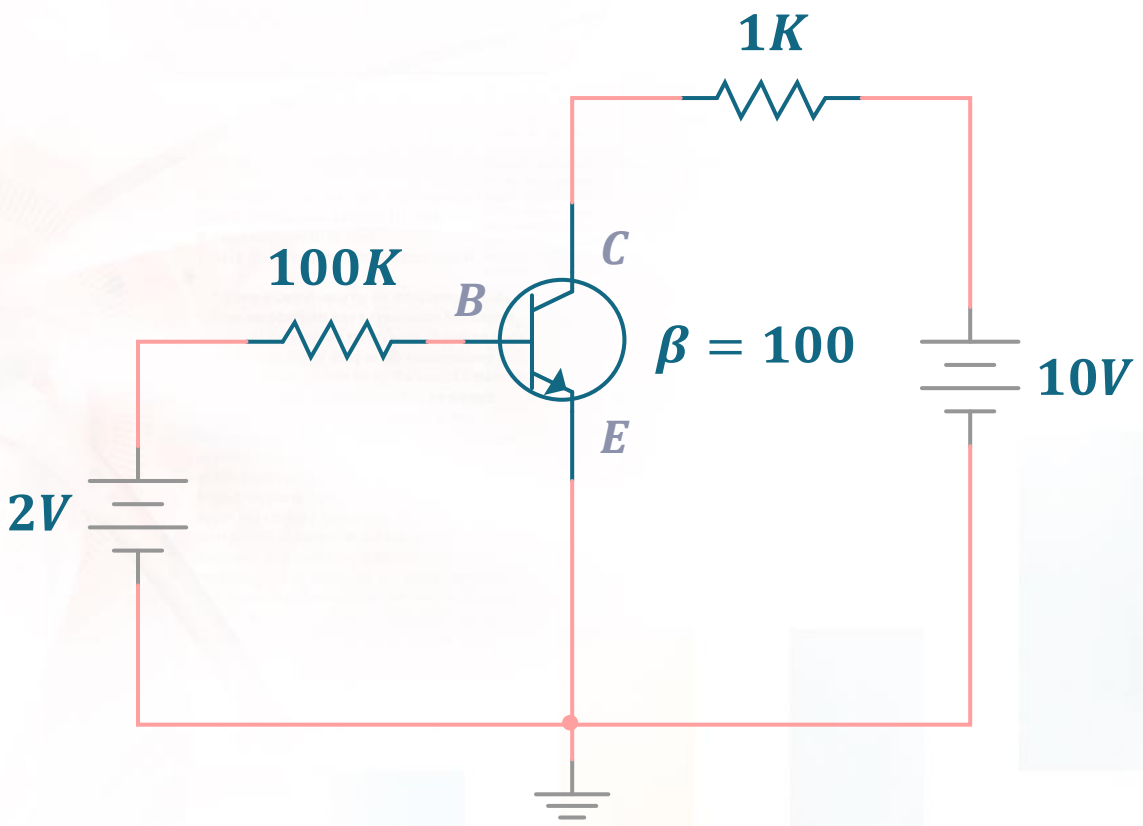
$$\beta_{dc} = \frac{i_C}{i_B}$$

Determine the parameters of the given transistor network.

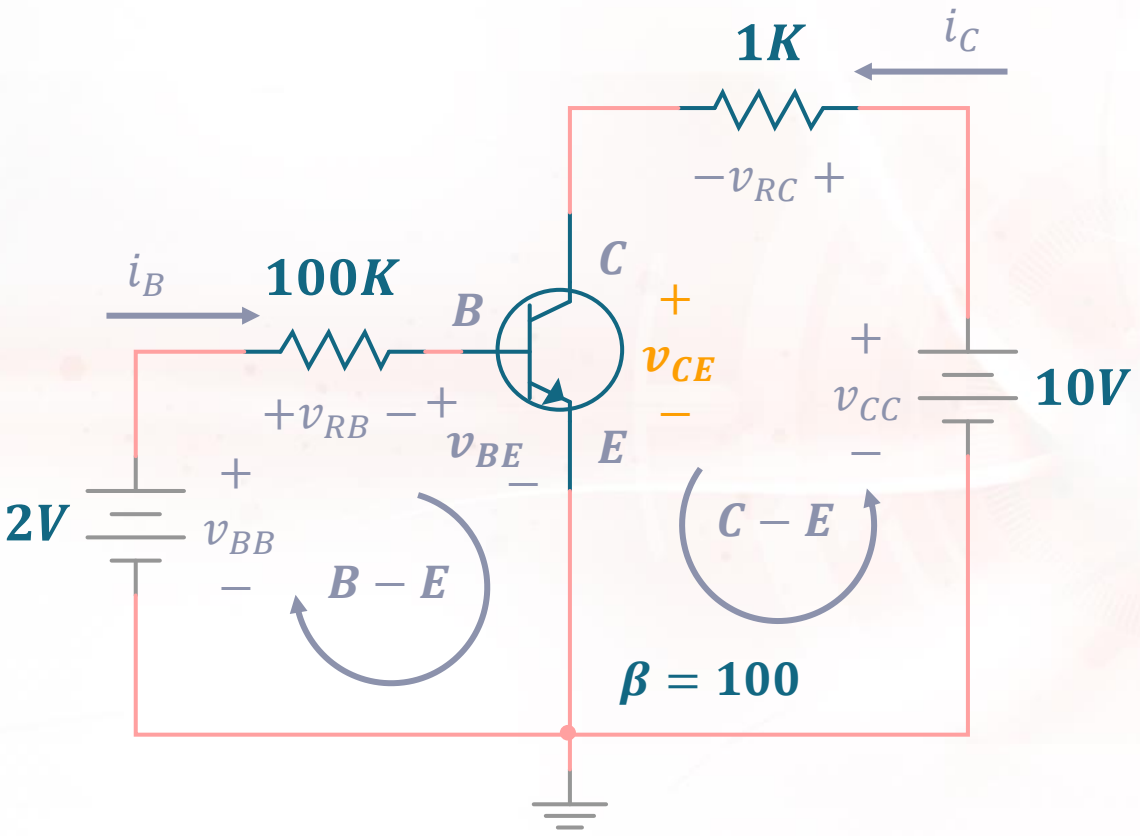
$i_B, i_C$, and $v_{CE}$

$i_c$ @ $v_{CE} = 0$

$v_{CE}$ @ $i_C = 0$



$i_{B-}$




KVL @ $B - E$,

$$-v_{BB} + v_{RB} + v_{BE} = 0$$

$$v_{RB} = v_{BB} - v_{BE}$$

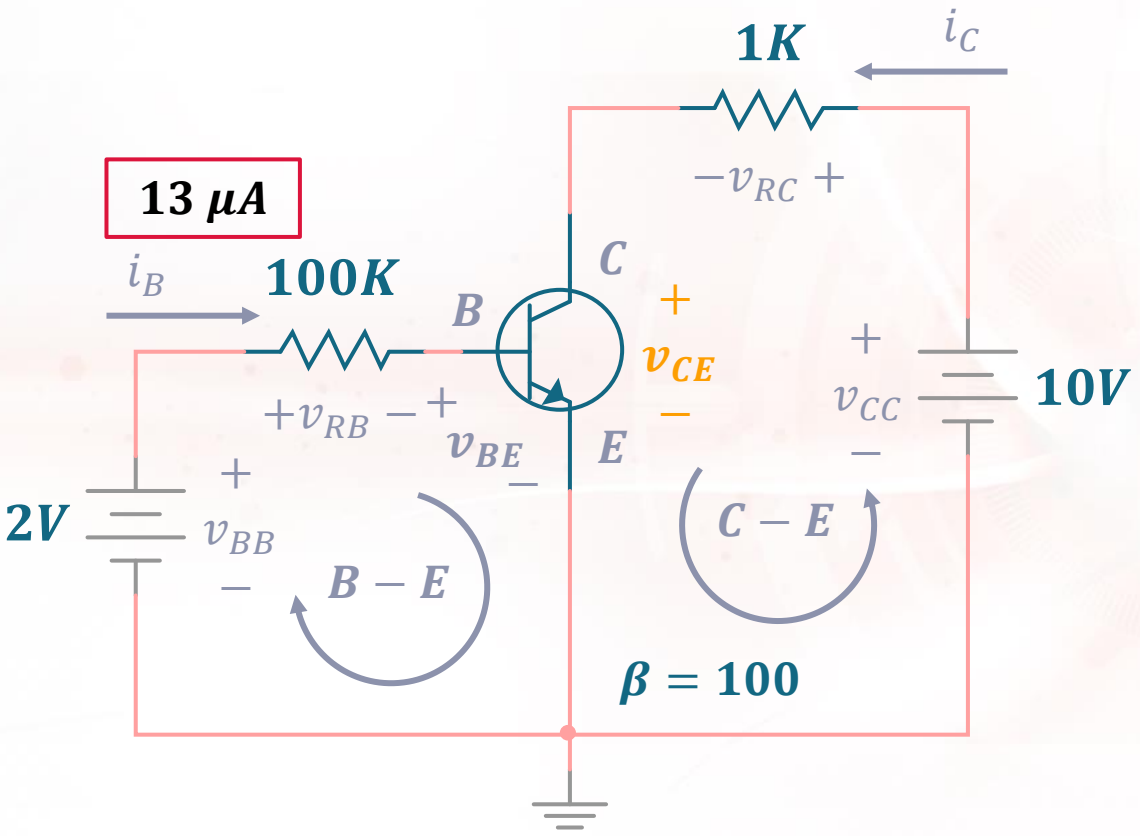
$$i_B R_B = v_{BB} - v_{BE}$$

$$i_B = \frac{v_{BB} - v_{BE}}{R_B}$$

$$i_B = \frac{2 - 0.7}{100K}$$

$$\boxed{i_B = 13\ \mu A}$$

*ans*




Current Gain,

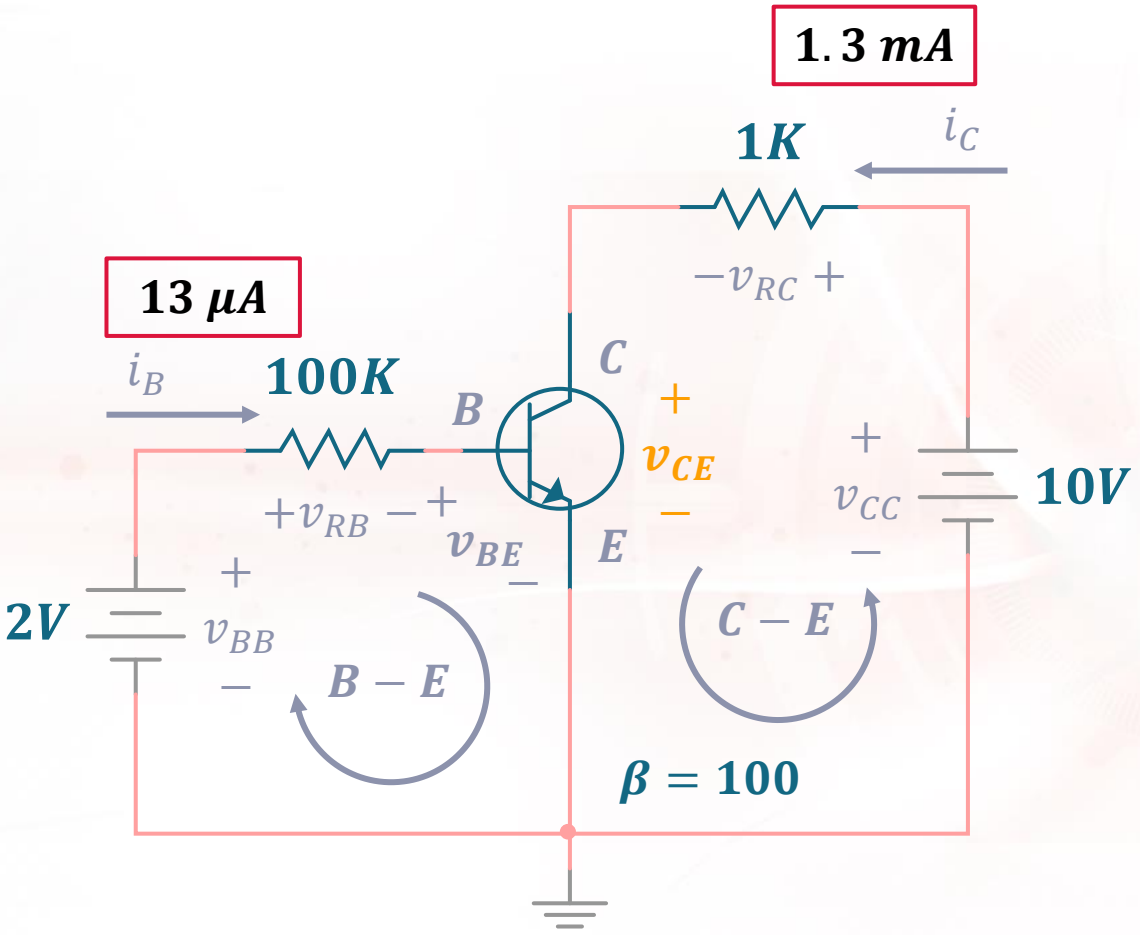
$$\beta = \frac{i_C}{i_B}$$

$$i_C = \beta i_B$$

$$i_C = 100(13\ uA)$$

$$\boxed{i_C = 1.3\ mA}$$

*ans*




KVL @ $C - E$,

$$-v_{CE} - v_{RC} + v_{CC} = 0$$

$$v_{CE} = v_{CC} - v_{RC}$$

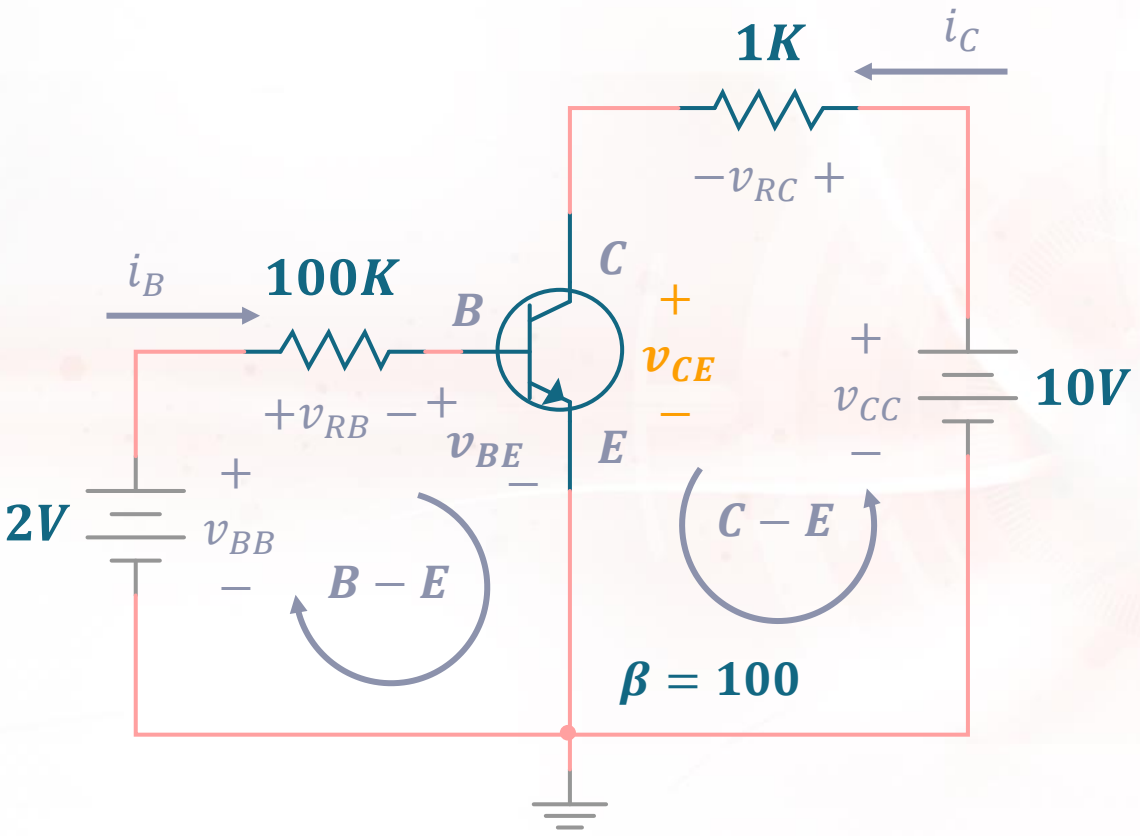
$$v_{CE} = v_{CC} - i_C R_C$$

$$v_{CE} = 10 - 1.3\ mA(1K)$$

$$\boxed{v_{CE} = 8.7\ V}$$

*ans*

# $v_{CE}$ @ $i_C = 0$




KVL @ $C - E$,

$$-v_{CE} - v_{RC} + v_{CC} = 0$$

$$v_{CE} = v_{CC} - v_{RC}$$

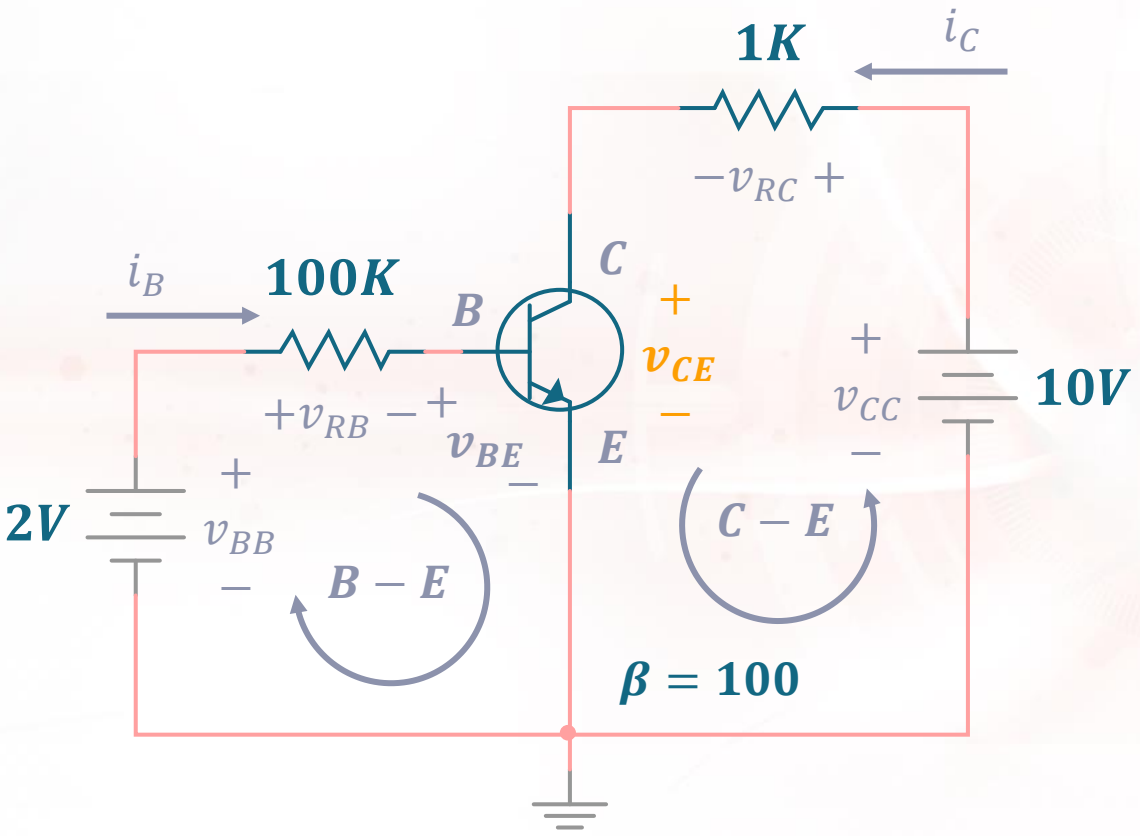
$$v_{CE} = v_{CC} - \cancelto{0}{i_C R_C}$$

$$v_{CE} = v_{CC}$$

$$\boxed{v_{CE(cutoff)} = 10\ V}$$

*ans*

# $i_c$ @ $v_{CE} = 0$




KVL @ $C - E$,

$$-v_{CE} - v_{RC} + v_{CC} = 0$$

$$\cancelto{0}{v_{CE}} = v_{CC} - v_{RC}$$

$$v_{RC} = v_{CC}$$

$$i_C R_C = v_{CC}$$

$$i_C = \frac{v_{CC}}{R_C}$$

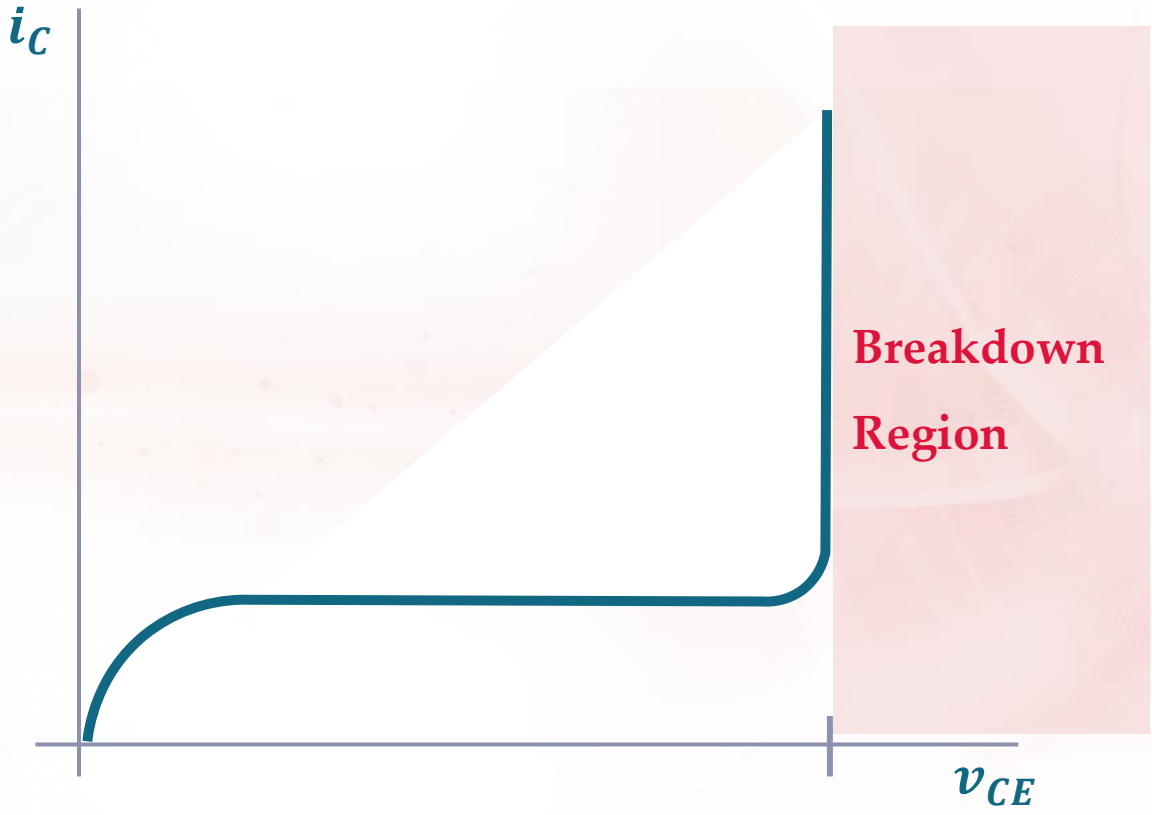
$$i_C = \frac{10}{1K}$$

$$\boxed{i_{C(sat)} = 10\ mA}$$

*ans*

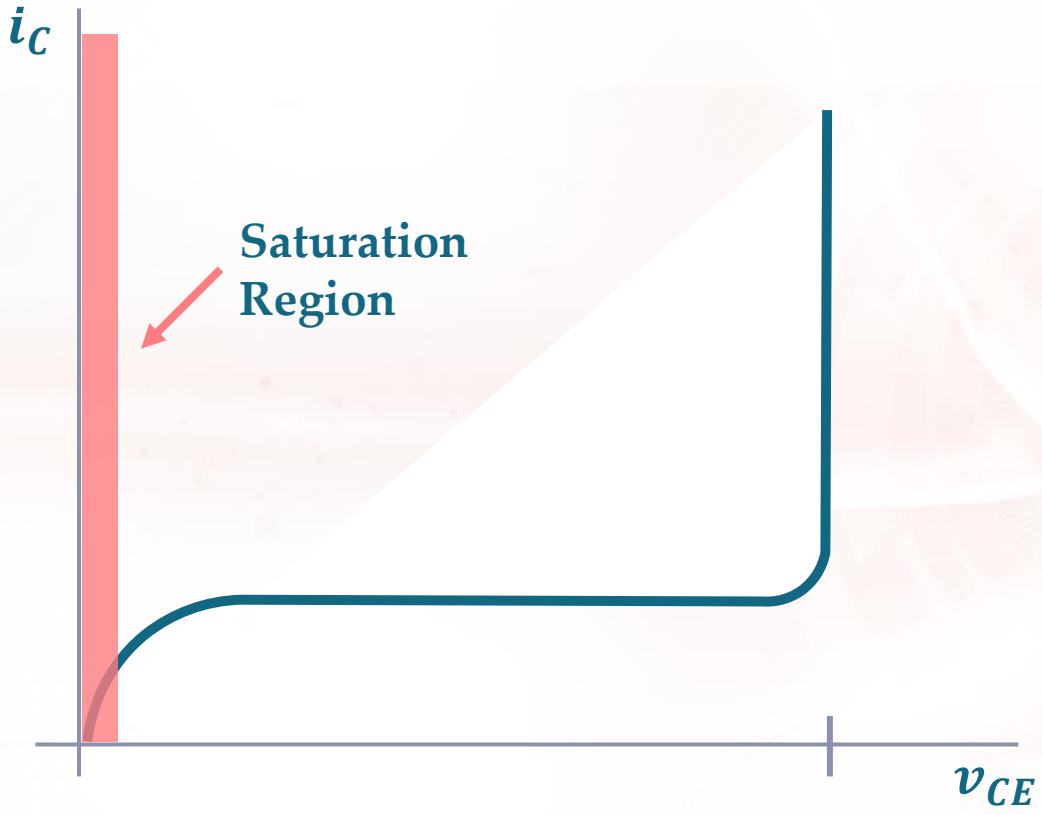
# REGIONS OF OPERATION

# BREAKDOWN REGION



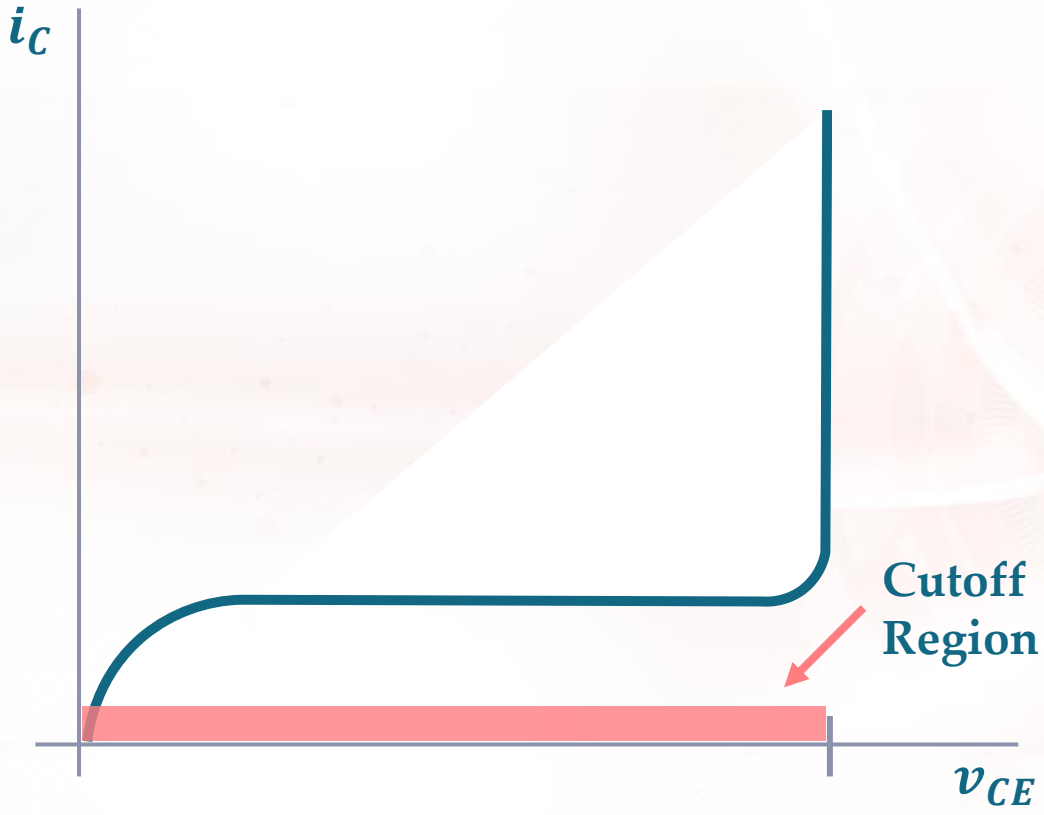

The transistor should never operate in this region because it will be **destroyed**

# SATURATION REGION



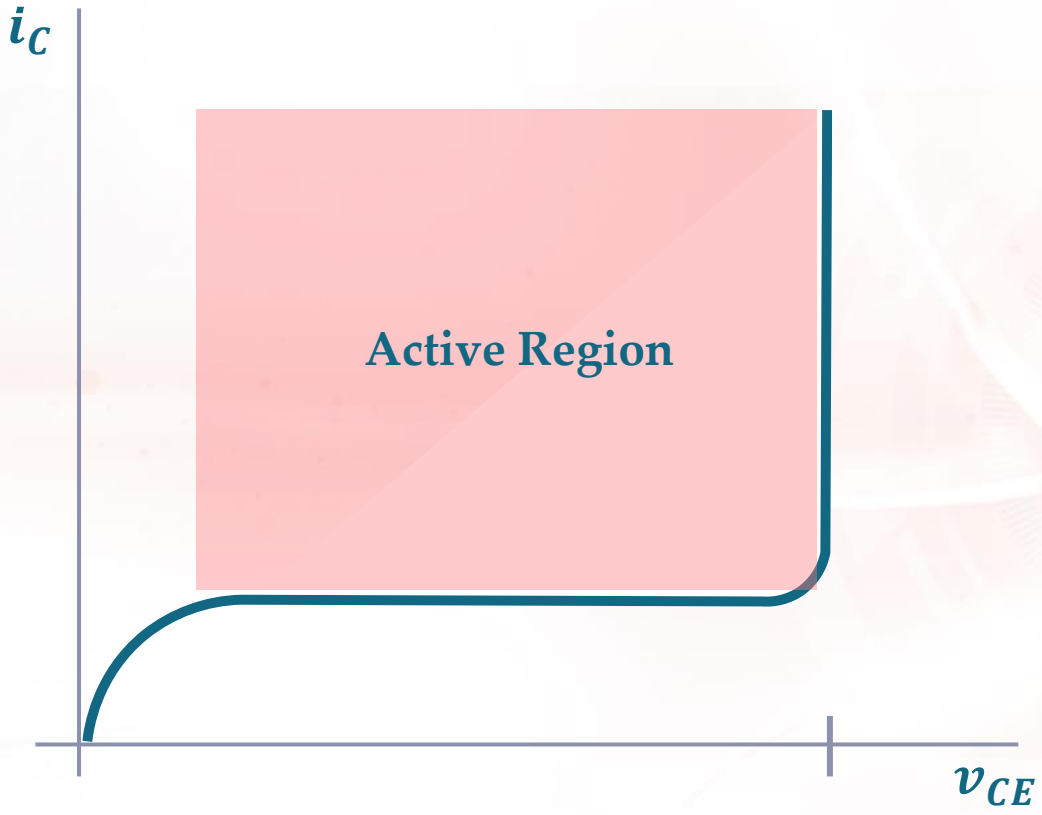

The early rising part of the curve, where $v_{CE}$ is between **0V** and few tenths of a volt

# CUTOFF REGION




A **small collector current** when the base current is zero

# ACTIVE REGION

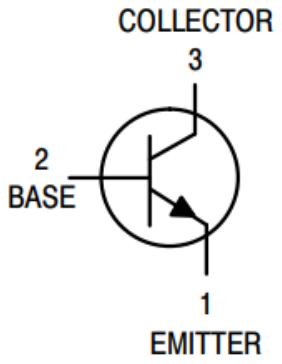


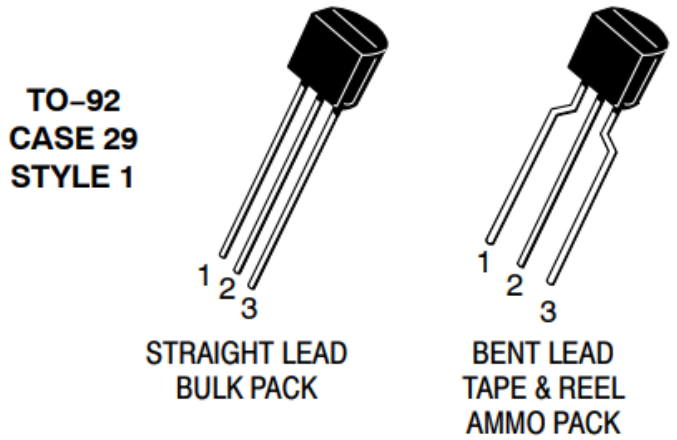

Collector current is **constant** in this region

# READING DATASHEET

## MAXIMUM RATINGS

| Rating | Symbol | Value | Unit |
|---|---|---|---|
| Collector – Emitter Voltage | $V_{CEO}$ | 40 | Vdc |
| Collector – Base Voltage | $V_{CBO}$ | 60 | Vdc |
| Emitter – Base Voltage | $V_{EBO}$ | 6.0 | Vdc |
| Collector Current – Continuous | $I_C$ | 200 | mAdc |
| Total Device Dissipation<br>@ $T_A$ = 25°C<br>Derate above 25°C | $P_D$ | <br>625<br>5.0 | <br>mW<br>mW/°C |
| Total Device Dissipation<br>@ $T_C$ = 25°C<br>Derate above 25°C | $P_D$ | <br>1.5<br>12 | <br>W<br>mW/°C |
| Operating and Storage Junction Temperature Range | $T_J$, $T_{stg}$ | –55 to +150 | °C |



**ELECTRICAL CHARACTERISTICS** ($T_A$ = 25°C unless otherwise noted)

| Characteristic | | Symbol | Min | Max | Unit |
|---|---|---|---|---|---|
| **ON CHARACTERISTICS** | | | | | |
| DC Current Gain (Note 2) | | $h_{FE}$ | | | – |
| ($I_C$ = 0.1 mAdc, $V_{CE}$ = 1.0 Vdc) | 2N3903 | | 20 | – | |
| | 2N3904 | | 40 | – | |
| ($I_C$ = 1.0 mAdc, $V_{CE}$ = 1.0 Vdc) | 2N3903 | | 35 | – | |
| | 2N3904 | | 70 | – | |
| ($I_C$ = 10 mAdc, $V_{CE}$ = 1.0 Vdc) | 2N3903 | | 50 | 150 | |
| | 2N3904 | | 100 | 300 | |
| ($I_C$ = 50 mAdc, $V_{CE}$ = 1.0 Vdc) | 2N3903 | | 30 | – | |
| | 2N3904 | | 60 | – | |
| ($I_C$ = 100 mAdc, $V_{CE}$ = 1.0 Vdc) | 2N3903 | | 15 | – | |
| | 2N3904 | | 30 | – | |
| Collector – Emitter Saturation Voltage (Note 2) | | $V_{CE(sat)}$ | | | Vdc |
| ($I_C$ = 10 mAdc, $I_B$ = 1.0 mAdc) | | | – | 0.2 | |
| ($I_C$ = 50 mAdc, $I_B$ = 5.0 mAdc | | | – | 0.3 | |
| Base – Emitter Saturation Voltage (Note 2) | | $V_{BE(sat)}$ | | | Vdc |
| ($I_C$ = 10 mAdc, $I_B$ = 1.0 mAdc) | | | 0.65 | 0.85 | |
| ($I_C$ = 50 mAdc, $I_B$ = 5.0 mAdc) | | | – | 0.95 | |

# LABORATORY